

USD/CRC · MONEX — Long/Short Quant Report (full history, net of real slippage)

2,663 sessions · 2014-12-08 → 2026-06-19 · execution cost 0.65 CRC/side (~13.7 bps) · long/short USD positions

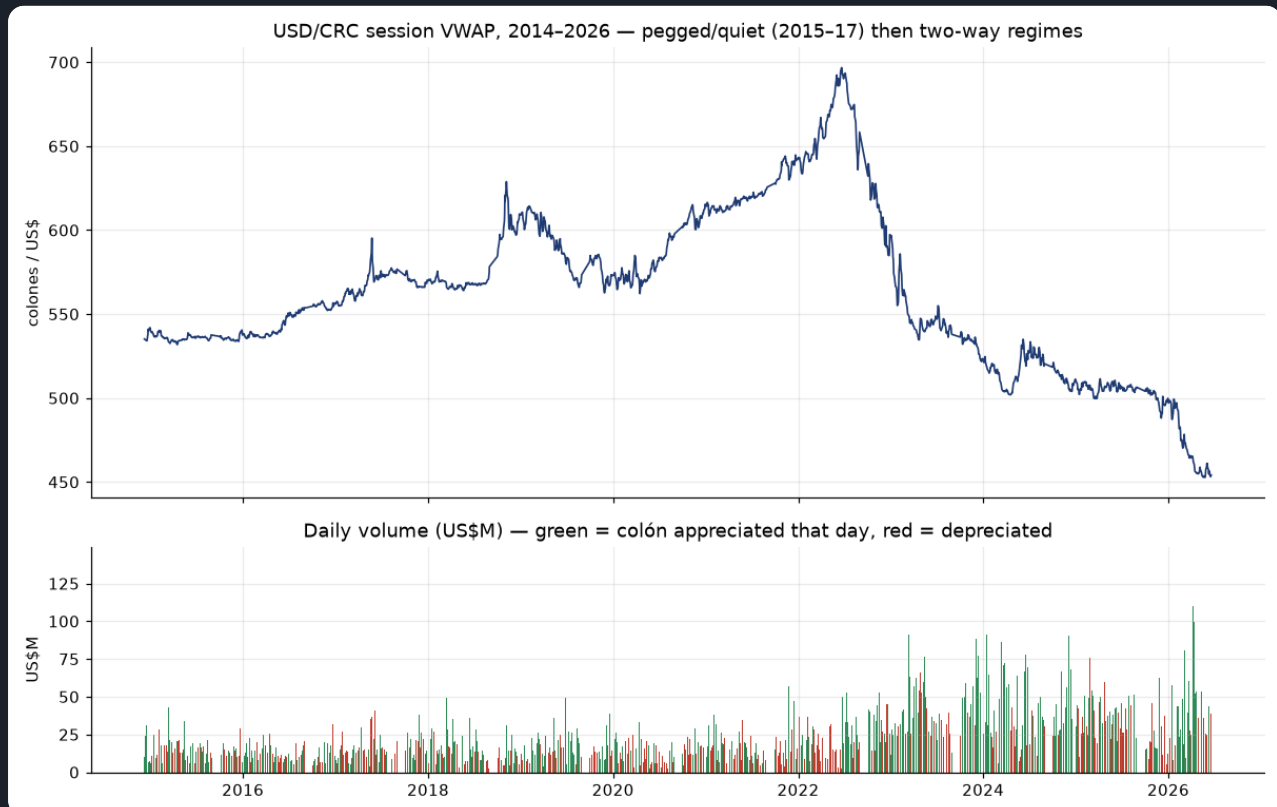
This revises the earlier 18-month study with **11.5 years of data** and your **real slippage of 0.65 CRC per side**. Both change the conclusion. The 18-month sample was a favourable window with an untradeable VWAP fill; on the full history and real costs, the high-frequency edge does not survive — but a **slow trend-following long/short** does, with a believable net Sharpe near 1.

- **REVISION** — the earlier "Sharpe ~5" was a mirage. It came from an 18-month window (the colón only became signal-rich after 2018) and from marking P&L at the VWAP, which you cannot actually trade at.
- **DEAD** — daily order-flow loses money net of your cost. The signal is *real* (gross next-day Sharpe 2.8, positive every year since 2017), but it flips position ~95.2×/yr and the ~5 bps/trade edge is smaller than your ~13.7 bps/side slippage → **net Sharpe -2.11**. It only pays if you are the liquidity *provider*, not the taker.
- **SURVIVES** — a slow trend-following long/short USD. Donchian-40 breakout: **net Sharpe 1.04**, 2.5 round-trips/yr, +452 bps/yr, max drawdown -12.9%. Both the long-USD and short-USD legs make money; permutation $p=0.0$.
- **CAVEAT** — it only earns when the colón trends. In-sample (2015–21, mostly the pegged years) net Sharpe was just 0.28; out-of-sample (2021–26, two-way regimes) 1.83. If the BCCR re-pins the rate, trend-following goes flat — it won't bleed, but it won't earn. Managed-float tail risk remains the dominant danger.

PART A · EXPLORE — 11.5 YEARS, MULTIPLE REGIMES

A1. Price & order flow across regimes

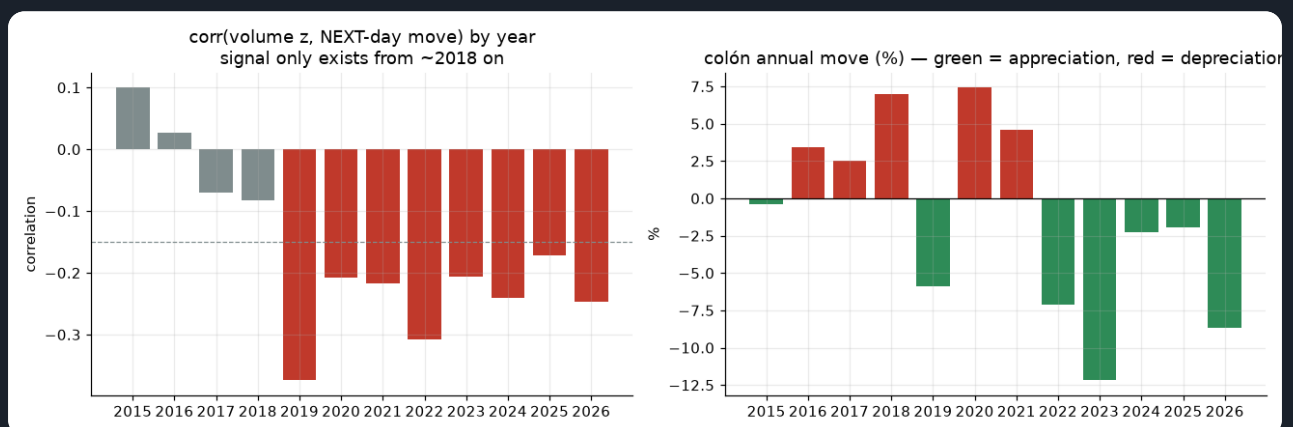
Session VWAP and signed daily volume, 2014–2026



The colón was effectively pinned/quiet in 2015–17, then traded two-way: depreciation spikes (2018, 2020, COVID) and a long appreciation 2022–26. A single strategy must cope with all of these.

A2. The signal is regime-dependent (the key overfitting lesson)

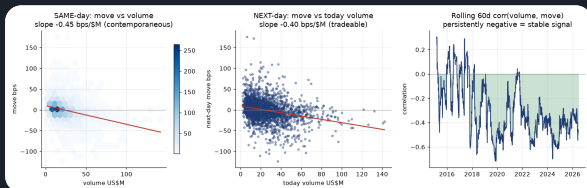
Volume → direction correlation by year, and the colón's annual move



Left: the volume → next-day-move correlation was ~0 in 2015–17 and only emerged from 2018 on. The original 18-month study sat entirely inside the favourable post-2018 regime — which is exactly how a real edge and an overfit one can look identical on a short sample. Right: the two-way moves a long/short strategy has to navigate.

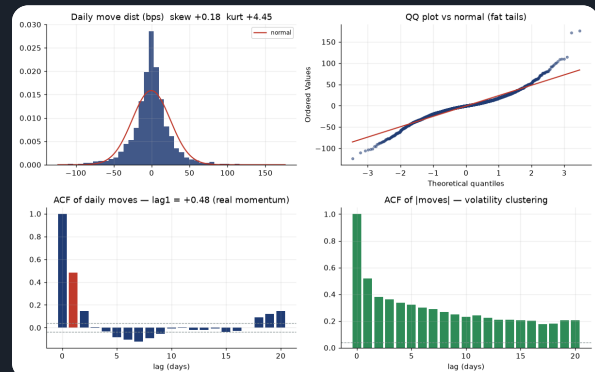
A3. The order-flow relationship & return character

Volume vs same-day & next-day move + rolling correlation



Heavy volume $\Rightarrow$ colón strength, contemporaneously and next-day — a genuine order-flow signal (USD supply). This is real; the problem is monetising it net of cost.

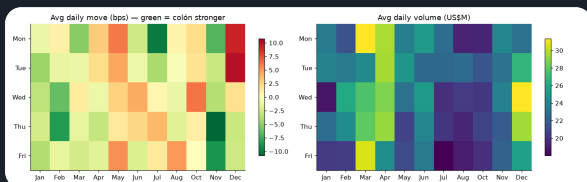
Return distribution, QQ, and autocorrelations



Fat tails and volatility clustering — the tails are what a managed-float blow-up would live in, and a short sample never shows them.

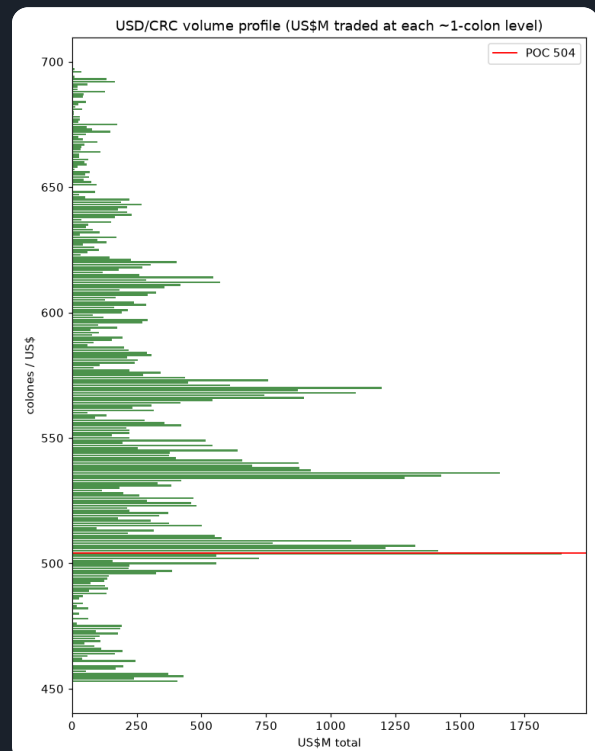
A4. Seasonality & volume profile (context overlays)

Day-of-week $\times$ month heatmaps



Calendar tilts in move and volume — a sizing overlay, not a standalone edge.

Volume at each price level

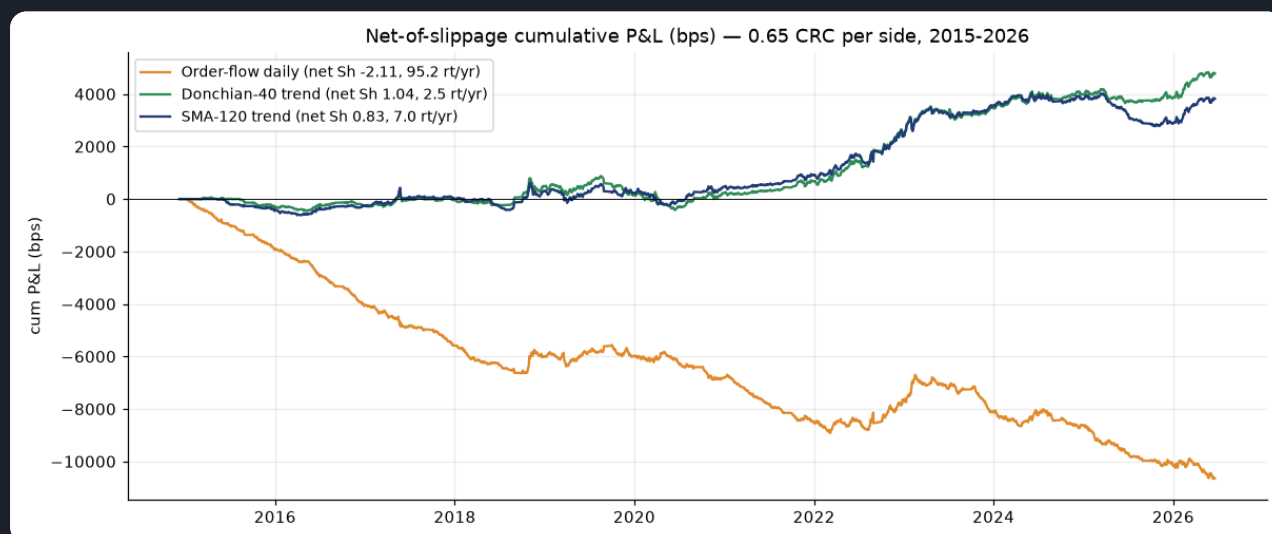


Where the market has traded over 11.5 years; congestion vs discovery zones.

B1. What 0.65 CRC/side does to each long/short strategy

LONG/SHORT STRATEGY	GROSS SH	NET SH	NET BPS/YR	ROUND-TRIPS/YR	MAX DD	VERDICT
Order-flow daily	2.8	-2.11	-1008	95.2	-106.6%	Dead — cost > edge
Donchian-40 trend	1.17	1.04	452	2.5	-12.9%	Survives
SMA-120 trend	1.21	0.83	361	7.0	-12.6%	Marginal

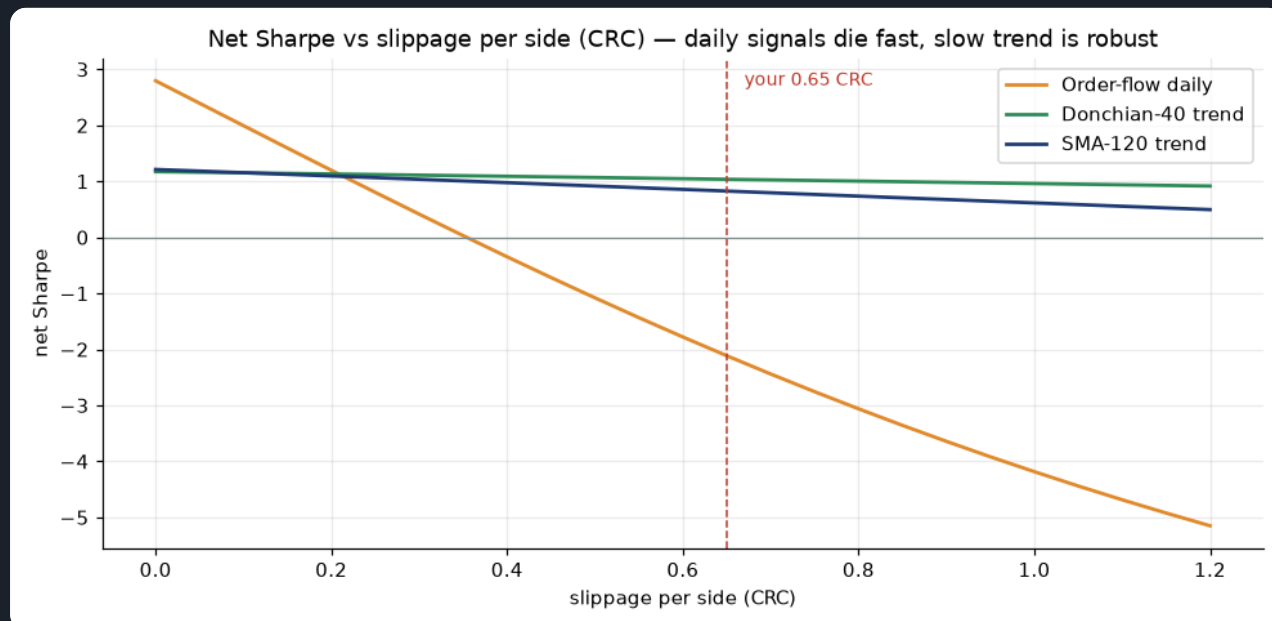
Net-of-slippage cumulative P&L — the whole story in one chart



Order-flow (orange) bleeds steadily to -10,000 bps purely on transaction cost. The trend strategies (green/navy) are flat through the 2015–21 quiet years, then earn +~4,000 bps in the trending era. Frequency, not signal quality, is what kills order-flow.

B2. Sensitivity to the slippage assumption

Net Sharpe vs slippage per side



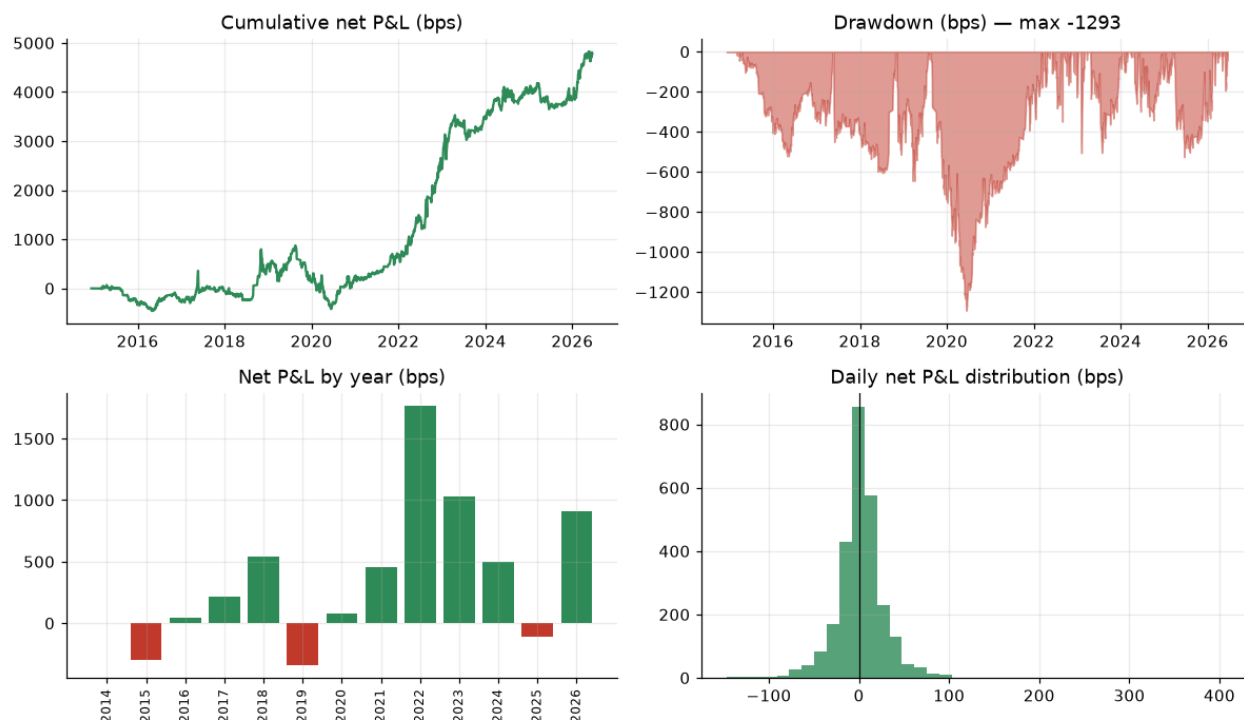
Daily order-flow crosses below zero almost immediately; the trend strategy stays positive well past your 0.65 CRC. The deployable edge is the one that barely cares about the exact cost.

PART C · THE DEPLOYABLE STRATEGY — TREND LONG/SHORT USD

C1. Net-of-cost tearsheet

Donchian-40 breakout long/short USD — net of 0.65 CRC slippage

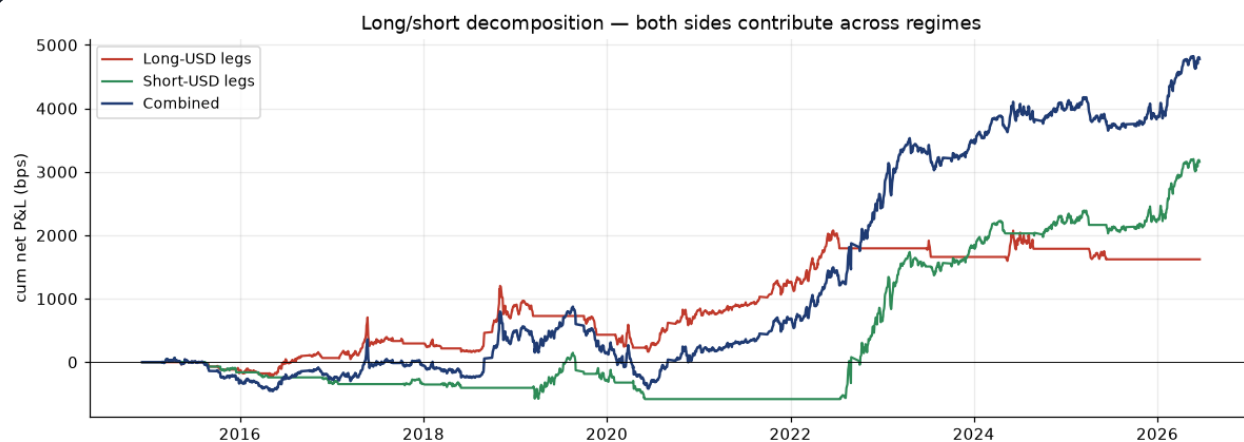
Donchian-40 long/short USD — NET of 0.65 CRC slippage (Sharpe 1.04 · 55.3% hit · 2.5 rt/yr · 452 bps/yr)



Equity, drawdown, P&L by year and the daily-return distribution. Flat 2015–21, then the bulk of the P&L in the trending 2022–26 years. Drawdowns are real (max -12.9%).

C2. It is genuinely two-way (long & short both work)

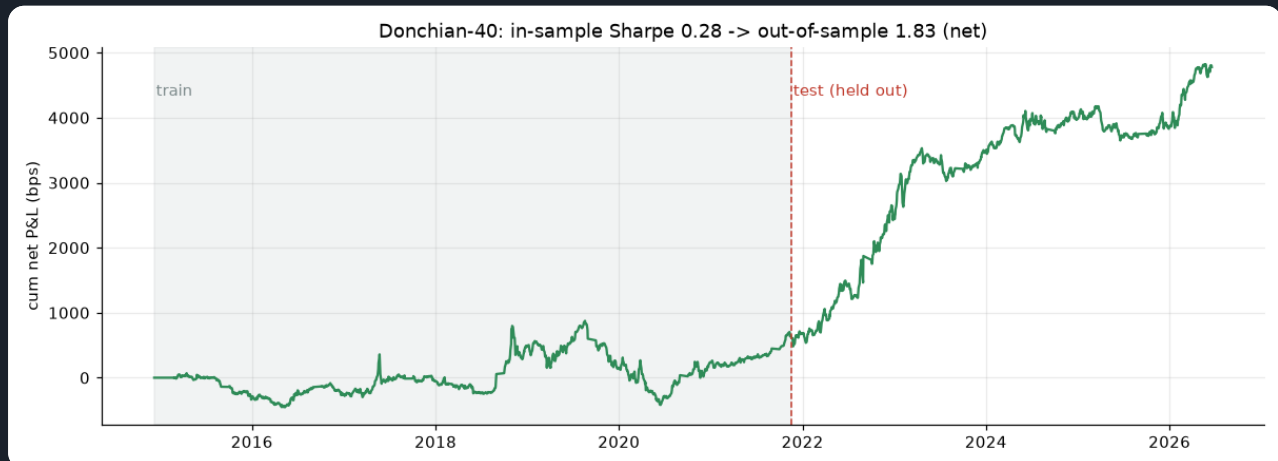
Long-USD vs short-USD leg contribution



Long-USD legs +1621 bps, short-USD legs +3158 bps. Both sides are positive across the sample, so this is a real long/short — not a one-directional bet dressed up.

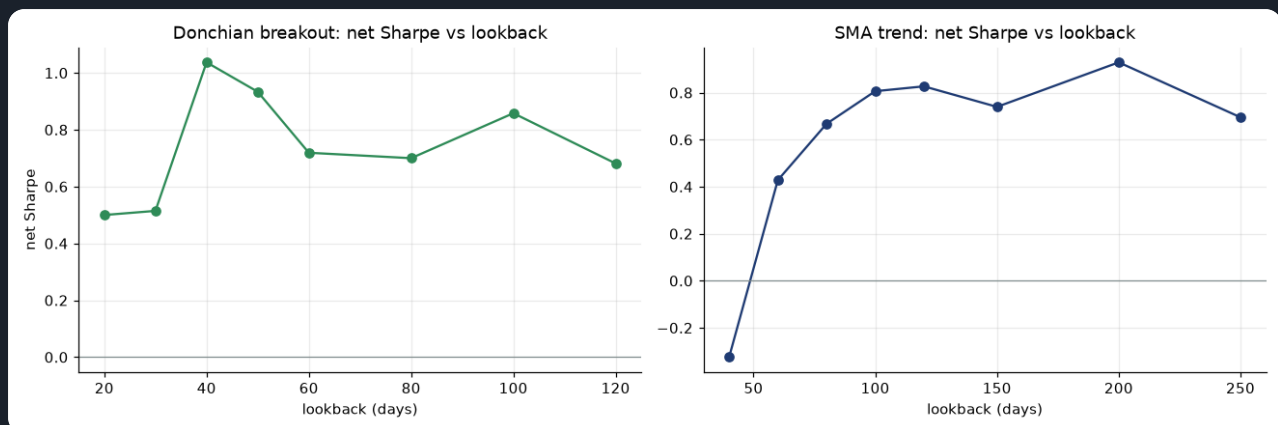
C3. Out-of-sample & robustness

In-sample (2015–21) vs out-of-sample (2021–26), net



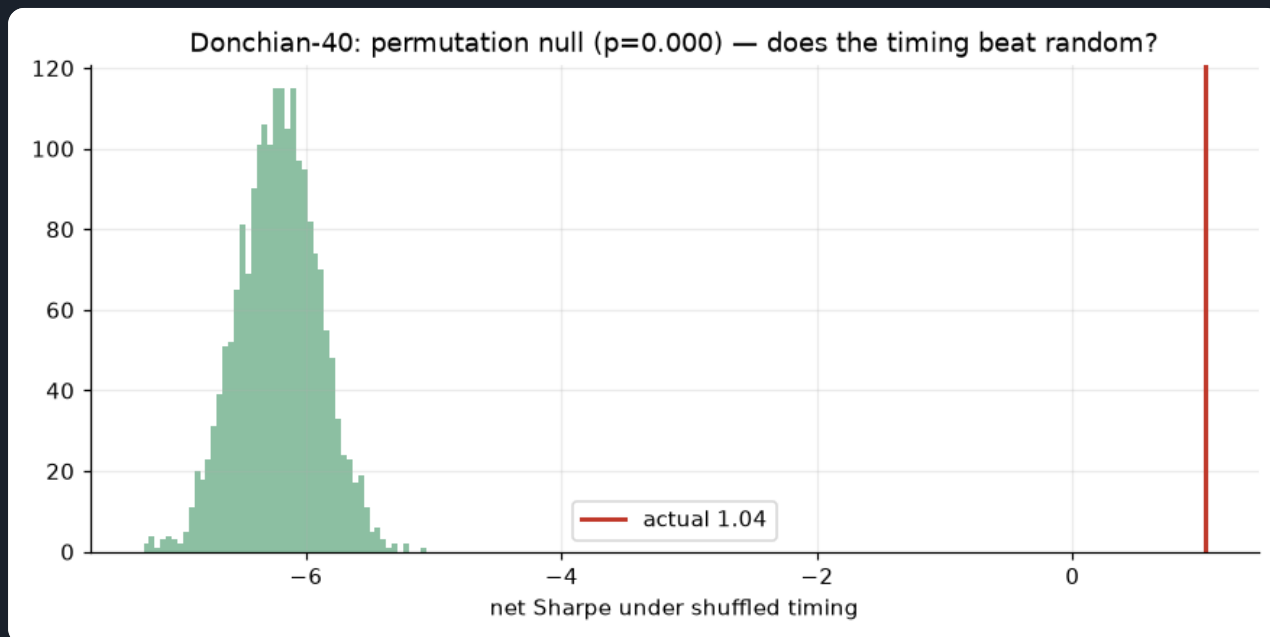
Net Sharpe 0.28 in-sample → 1.83 out-of-sample. The OOS period is stronger because it contained the trends; the honest read is 'pays when the colón moves'.

Net Sharpe vs lookback (Donchian & SMA)



Positive across a broad band of lookbacks (Donchian 0.5–1.04), so it isn't a single lucky parameter — but it is not a high-Sharpe machine either.

Permutation null (net)



Shuffling the timing of the same positions destroys the edge ($p=0.0$), so the entry/exit timing carries genuine information.

PART D · VERDICT & DEPLOYMENT

Is it worth pursuing?

As a standalone speculative book: marginal. Net Sharpe ~1 with ~13% drawdowns and a known tail risk is not compelling on its own, and you need an instrument that lets you be long *and* short USD/CRC at 0.65 CRC slippage.

As a treasury / hedging overlay (most likely your case): yes, useful. If you already run USD/CRC exposure, a slow trend filter that tells you when to be long vs short USD — trading only a few times a year — is a cheap, robust improvement over a static hedge, and it survived 11.5 years and real costs.

Do NOT deploy the daily order-flow signal as a taker. It is real but you will pay it all away in slippage. The only way to monetise it is as a market-maker / internaliser capturing the spread you are currently paying.

Dominant risk: the BCCR. A re-peg flattens the trend edge; a policy break / intervention is the fat tail this sample can't price. Size for survival, keep the lookback slow, and re-estimate continuously.

Caveats. P&L is net of your stated 0.65 CRC/side slippage but gross of financing/borrow and any market impact; the colón is not freely shortable for all participants. Tests assume you can hold a signed USD position at MONEX. Your slippage example described price *improvement* (buy below / sell above spot); this report models the economically standard *adverse* 0.65 CRC — tell me if you meant otherwise and the cost layer flips. The "best board offers" block in the export was empty, so VWAP / close are the price proxies.

Reproducible: `parse_monex` → `analyze` → `eda` → `strategies` → `backtest` → `build_report`, then `weasyprint` for the PDF. Strategy numbers are read live from `backtest_results.json`.